

Robust multi-objective optimization of roundwood timber bridges: Structural performance and parametric design relationships

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Abstract

The design of timber bridges involves multiple design variables and conflicting code requirements, and is traditionally carried out through iterative procedures that depend on the designer's experience, without a systematic investigation of how span, strength class and dimensional variability interact in the structural behaviour. This study proposes a parametric robust multi-objective optimization framework for roundwood timber bridges, with two conflicting objective functions, the timber volume and the utilization of the deflection limit, solved by NSGA-II under the ultimate and serviceability limit state constraints of ABNT NBR 7190:2022. Robustness is incorporated by keeping the objectives at the nominal point and aggregating the constraints by the worst case over a deterministic five-point deviation grid $\pm\rho$ ($\rho = 5\%$) applied to the girder diameter. The framework was applied to 20 configurations combining four spans (3.0 to 6.0 m) and the five strength classes D20 to D60 under the TB-240 design vehicle. All configurations converged to feasible solutions, with timber consumption between 1.75 and 9.98 m³. Consumption grows with span according to a power law with exponent between 1.64 and 1.73, and the first class upgrade, from D20 to D30, accounts for 41 to 48% of the total saving available up to D60. Bending governs all configurations, fifteen through the deck and five through the girder, and none is governed by deflection, although deflection approaches its limit in the most severe cases. Robustness carries a real and monotonic cost in volume, from +6.0% ($\rho = 5\%$) to +26.9% ($\rho = 20\%$). The results were condensed into preliminary design charts and a pre-sizing equation ($R^2 = 0.999$), valid within the investigated domain, quantifying how span, strength class and dimensional variability control material demand and the governing limit states.

Keywords: roundwood timber bridges, multi-objective optimization, robust design, parametric design, structural optimization, timber structures.

Nomenclature

A	Cross-sectional area (m ²)
b_w	Width of the deck plank cross-section (m)
CIV	Vertical impact coefficient
d	Diameter of the circular girder (m)
esp_{long}	Spacing between girders (m)
esp_{tab}	Spacing between deck planks (m)
$E_{0,m}$	Mean modulus of elasticity in bending (kPa)

$f_{c0,k}$	Characteristic compressive strength parallel to the grain (MPa)
$f_{m,d}$	Design bending strength (kPa)
$f_{v0,d}$	Design shear strength parallel to the grain (kPa)
f_1, f_2	Objective functions, total timber volume and utilization of the deflection limit
$f_m(\mathbf{x})$	m -th objective function
$g_j(\mathbf{x})$	j -th inequality constraint
h	Height of the deck plank cross-section (m)
I	Second moment of area of the cross-section (m ⁴)
k_{mod}	Modification factor
L	Design span of the girder (m)
M_{gk}	Characteristic bending moment due to permanent actions (kN·m)
M_{qk}	Characteristic bending moment due to variable actions (kN·m)
M_{sd}	Design bending moment (kN·m)
P	Wheel load of the design vehicle (kN)
a	Axle spacing of the design vehicle (m)
Q_{sd}	Design shear force (kN)
V_{total}	Total timber volume of the bridge (m ³)
W	Section modulus of the cross-section (m ³)
$\mathbf{x}$	Vector of design variables
Λ, λ	Deterministic diameter perturbation grid and its generic element
γ_g, γ_q	Partial factors for permanent and variable actions
γ_{wf}, γ_{wc}	Material partial factors (bending and shear)
δ_{tot}	Total midspan deflection (m)
δ_{lim}	Allowable deflection limit (m)
φ	Creep coefficient
ψ_2	Quasi-permanent combination factor
ρ	Perturbation factor of the design variable (robustness)
N_c	Number of points of the diameter perturbation grid in the robust evaluation
HV	Hypervolume indicator of the approximated front
ST_i	Total-effect Sobol index of variable i

1. Introduction

Timber has accompanied the history of construction since the earliest structural systems and remains relevant owing to its eco-efficiency, the low energy consumption of its production and its high strength-to-weight ratio (Fraga et al., 2025). In buildings, the embodied energy of timber structures is, on average, 28% and 47% lower than that of equivalent concrete and steel structures, respectively (Schenk & Amiri, 2022). Its combination with concrete and steel enables composite structural systems that exploit the properties of each material, widening the range of applications of timber and improving structural performance (Pastori et al., 2022). In view of the global decarbonization targets of the construction sector (Li et al., 2025), timber re-emerges as a promising structural alternative for small and medium-span bridges, particularly on low-traffic rural roads, where spans are short and the demand for economical and sustainable solutions is high (Milani et al., 2020). In many of these contexts, concrete and steel solutions are hindered by the lack of local production and by the logistical difficulty of transporting these materials to

the site, which makes timber, frequently available in the region itself, the most viable structural alternative.

In this scenario, computational structural optimization has become established as a tool to reconcile performance and environmental targets, and has already been employed in several fields of engineering design. Moraes et al. (2022), for instance, employed metaheuristic optimization to reduce the CO₂ emissions associated with the design of precast prestressed concrete beams, whereas Fraga et al. (2025) applied optimization concepts to the design of timber members in Fink trusses.

Historically, the structural design process has evolved from manual calculations and semi-automated computer-aided drafting procedures towards approaches that integrate parametric modelling, automated structural analysis and data-driven optimization techniques. This paradigm has already been applied successfully to bridge design. By coupling three-dimensional parametric modelling with global and local optimization routines, Moraes and Buttignol (2025) automated the design of prestressed concrete girders for highway bridges. At industrial scale, Rajic (2026) developed a parametric toolbox that automates the generation of finite element models and the code checks of prestressed concrete bridges, successfully applied to large infrastructure projects. Platforms of this kind have also been employed for the structural assessment and diagnosis of bridges (Alseid & Seo, 2025; Garbowski et al., 2023; Miluccio et al., 2026).

This literature, however, focuses predominantly on concrete bridges. Parametric frameworks for timber structures, such as that of Jiang et al. (2023), are restricted to building components. Their extension to roundwood timber bridges remains little explored, especially regarding the dimensional variability of the members and its influence on the structural behaviour. The available studies on timber bridges focus mostly on field surveys and pre-sizing tables based on manual checks, such as that of Criado et al. (2024), or on dynamic testing and numerical modelling of existing bridges, such as that of Bergenudd et al. (2023), without incorporating parametric modelling, structural optimization or robustness treatment. A recent systematic review on the structural optimization of timber components further points out that the lack of open-source codes is one of the main barriers to transferring this research to practical applications (Belz & Kromoser, 2025). The literature on bridge optimization therefore remains dominated by other materials, and for roundwood timber highway bridges the integrated investigation of geometry, strength properties, dimensional variability, material efficiency and governing limit states is still limited. This is the gap that the parametric robust optimization framework proposed in this work seeks to fill, with the open-access computational platform serving as an instrument for implementing and transferring the results. In this paradigm, design ceases to be a linear procedure and is treated as an iterative and integrated process, in which the geometric model, the internal force analysis and the design decision are coupled within a single optimization routine. This transition reduces design time, improves the quality of the solutions and allows the space of feasible alternatives to be explored systematically.

In this context, the design of timber bridges can be formulated as an optimization problem in which conflicting objectives must be considered simultaneously, such as minimizing material consumption, satisfying the ultimate and serviceability limit states and ensuring the overall structural performance. Metaheuristic-based multi-objective optimization emerges as a suitable tool to handle this complexity, since it yields a set of efficient solutions represented by a Pareto front (Deb, 2009), giving the designer greater flexibility in decision-making and favouring the

rational use of resources.

However, deterministic optimization frequently leads to solutions located exactly on the constraint boundary, where small variations in the design variables, arising from manufacturing tolerances, the dimensional variability of roundwood or assembly inaccuracies, may lead to the violation of safety criteria. This aspect is particularly critical for roundwood, whose diameter exhibits significant natural variability and whose load-bearing capacity is directly conditioned by this dimensional variability (Morgado et al., 2017). It is therefore necessary to incorporate the concept of robustness into the optimization process, so that the solutions obtained remain safe even under deviations around their nominal values.

1.1 Objectives and contributions

This work proposes a parametric robust multi-objective optimization framework for the design of roundwood timber bridges, in which the parametric modelling of the structural members is coupled to an evolutionary search routine, in accordance with the criteria of NBR 7190-1 (2022). Unlike approaches that design the deck and the girder in sequential and independent steps, the formulation adopted here couples both members in a single evaluation, detailed in Section 4.

The specific objectives of this work are as follows.

- (i) To formulate the design of the structural members of timber bridges as a multi-objective optimization problem, defining objective functions and constraints consistent with the safety and performance criteria of the code.
- (ii) To implement the NSGA-II algorithm coupled to the parametric structural model, ensuring the convergence and diversity of the solutions.
- (iii) To incorporate robustness analysis into the search process, evaluating the effect of a 5% deviation applied to the nominal girder diameter on the constraints.
- (iv) To investigate the parametric matrix of single-lane rural road bridges, covering four design spans and the five timber strength classes of the code, in order to identify the trends in material consumption, structural performance and governing limit states across the investigated domain.

The main contributions of this work fall into three categories, namely methodological, structural and applied. To the best of the authors' knowledge, the methodological contribution is the first approach to integrate parametric modelling, multi-objective optimization and robustness analysis in the design of roundwood timber bridges. The structural contribution stems from the investigation of the parametric matrix, which identifies the relationships between span, strength class, dimensional variability, material consumption and governing limit states. The applied contribution results from the evaluated case matrix, which yields preliminary design charts and a closed-form pre-sizing equation, made available through an open-access computational platform that implements and transfers these results to the designer.

2. Optimization

In this work, the design of the girders and deck planks is treated as a sizing optimization problem (Bendsøe & Sigmund, 2004), in which the topology of the structural system, that is, the number and arrangement of the members, is defined *a priori* by the designer, and the algorithm searches for the optimal dimensions of these members within the prescribed bounds.

2.1 Background on bridge optimization

The application of optimization techniques to bridge design has historically concentrated on concrete and steel bridges and predates the widespread use of evolutionary algorithms. It evolved from deterministic procedures aimed at cost minimization (Aguilar et al., 1973; Brar et al., 1984) to formulations that combine multiple decision levels, from the component to the structural configuration, and multiple performance criteria (Cohn & Lounis, 1994; Simões & Negrão, 1994). The adoption of metaheuristics extended this formulation to problems with numerous discrete variables and coupled structural checks (García-Segura & Yepes, 2016; Martí et al., 2013), subsequently incorporating reliability targets over the service life (García-Segura et al., 2017) and, more recently, three-dimensional modelling coupled with local search (Moraes & Buttignol, 2025).

For timber, the precedents focus on cable-stayed glued laminated timber footbridges, a structural system distinct from the roundwood highway bridges studied in this work. Negrão (1999) addressed cost minimization by varying member dimensions and material properties, and Simões and Negrão (2005) formulated a reliability-based optimum design procedure integrating finite element analysis, sensitivity analysis and a multi-criteria formulation of cost, stresses, displacements and reliability. These precedents show that both the search for economical solutions and the consideration of uncertainties are already part of timber bridge research, but without the joint treatment of the dimensional variability of roundwood and a multi-objective formulation, which is the focus of this work.

The robustness treatment adopted here, detailed in Section 2.4, evaluates the response of the solutions to prescribed dimensional variations and does not, by itself, correspond to imposing a probability of failure or a target reliability index. The multi-objective formulation used is presented next.

2.2 Multi-objective optimization framework

A multi-objective optimization problem involves the simultaneous minimization and/or maximization of multiple objective functions, subject to a set of constraints that define the feasibility of the candidate solutions. Unlike single-objective optimization, this formulation does not lead to a single global optimum. Instead, it results in a set of Pareto-optimal solutions, each representing a distinct trade-off between competing performance criteria.

The general mathematical formulation of a multi-objective optimization problem can be expressed as in Eq. (1).

$$\begin{cases} f_m(\mathbf{x}), & m = 1, 2, \dots, M; \\ g_j(\mathbf{x}) \leq 0, & j = 1, 2, \dots, J; \\ h_k(\mathbf{x}) = 0, & k = 1, 2, \dots, K; \\ x_i^{(L)} \leq x_i \leq x_i^{(U)}, & i = 1, 2, \dots, n. \end{cases} \quad (1)$$

In this formulation, the vector of objective functions is defined by Eq. (2).

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x}))^T \quad (2)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ is the n -dimensional vector of design variables. The feasible design space is bounded by the inequality constraints $g_j(\mathbf{x})$, the equality constraints $h_k(\mathbf{x})$ and the side constraints defined by the lower and upper bounds $x_i^{(L)}$ and $x_i^{(U)}$, which together

establish the admissible domain (D) (Deb, 2009).

Objectives originally conceived as maximization problems are converted into equivalent minimization problems by multiplying the corresponding objective functions by -1 , ensuring a unified computational structure.

2.3 NSGA-II optimization algorithm

Developed by Deb (2009), this evolutionary algorithm has become established in the literature for its ability to efficiently approximate the Pareto-optimal set. NSGA-II preserves the vector nature of the multi-objective problem and searches for a diverse set of non-dominated solutions without resorting to scalarization (weighted sum) of the objective functions. The core mechanism of the algorithm is fast non-dominated sorting. At each generation, the whole population is evaluated and sorted into successive non-dominated fronts $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_k$. The first front ($\mathcal{F}_1$) gathers the currently optimal solutions of that generation and constitutes the current approximation of the Pareto front. The next front ($\mathcal{F}_2$) contains the solutions that would be non-dominated if the individuals of $\mathcal{F}_1$ were disregarded, and so on. To prevent premature convergence to a single region and to maintain a uniform spread along the front, the algorithm uses the crowding distance operator. This metric estimates the density of solutions around a given individual by computing the hypervolume formed by its nearest neighbours. For a problem with M objective functions, the crowding distance d_i of individual i is computed by Eq. (3).

$$d_i = \sum_{m=1}^M \frac{f_m(\mathbf{x}_{i+1}) - f_m(\mathbf{x}_{i-1})}{f_m^{\max} - f_m^{\min}} \quad (3)$$

NSGA-II employs a strongly elitist strategy. At each cycle, the parent and offspring populations are merged and sorted first by Pareto rank and, in case of a tie within the same front, by the largest crowding distance. Algorithm 1 summarizes the evolutionary logic and the flow of structural evaluations, incorporating the protection against uncertainty described in Section 2.4.

The evolution of the generations relies on genetic operators suited to continuous variables. The combined use of simulated binary crossover (SBX) and polynomial mutation (PM) refines local solutions in the search space, while the systematic elimination of duplicates preserves the genetic diversity of the population and prevents premature convergence. The configuration parameters adopted in all runs of this work are listed in Table 1.

Table 1: NSGA-II configuration parameters.

Parameter	Value
Population size, N_{pop}	50
Number of generations, N_{gen}	300
Initial sampling	Uniform random
Crossover	SBX, $p_c = 0.90$, $\eta_c = 15$
Mutation	Polynomial, $\eta_m = 20$
Duplicate elimination	Yes
Random seed	1
Number of robustness grid points, N_c	5
Robustness deviation, ρ	5%

2.4 Robustness in optimization

In structural engineering, purely deterministic optimization tends to place the optimal solutions exactly on the mathematical boundary of the active constraints. This characteristic makes

Algorithm 1 Robust multi-objective optimization via NSGA-II

Require: Bounds $[x_i^{(L)}, x_i^{(U)}]$, population size N_{pop} , generations N_{gen} , robustness deviation ρ , robustness grid $\Lambda = \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$

Ensure: Efficient front $\mathcal{F}_1$

- 1: Initialize population P_0 by random sampling in $[x_i^{(L)}, x_i^{(U)}]$
- 2: **for** each generation $t = 0$ to $N_{gen} - 1$ **do**
- 3: **for** each individual $\mathbf{x} = (d, b_w, h, esp_{long}, esp_{tab}) \in P_t$ **do**
- 4: $(\mathbf{F}, \cdot) \leftarrow \text{EVALUATEMODEL}(\mathbf{x})$ $\triangleright$ objectives at the nominal point, no aggregation
- 5: $\mathbf{G} \leftarrow -\infty$
- 6: **for** each $\lambda \in \Lambda$ **do** $\triangleright$ robustness grid, applied to d only
- 7: $\tilde{d} \leftarrow d(1 + \rho\lambda)$
- 8: $\tilde{\mathbf{x}} \leftarrow (\tilde{d}, b_w, h, esp_{long}, esp_{tab})$
- 9: $(\cdot, \mathbf{g}) \leftarrow \text{EVALUATEMODEL}(\tilde{\mathbf{x}})$
- 10: $\mathbf{G} \leftarrow \max(\mathbf{G}, \mathbf{g})$ $\triangleright$ component-wise, worst case
- 11: **end for**
- 12: **end for**
- 13: $Q_t \leftarrow$ tournament selection, SBX crossover and polynomial mutation on P_t
- 14: $R_t \leftarrow P_t \cup Q_t$
- 15: Sort R_t into non-dominated fronts $\{\mathcal{F}_1, \mathcal{F}_2, \dots\}$
- 16: $P_{t+1} \leftarrow$ best N_{pop} individuals by Pareto rank and crowding distance
- 17: **end for**
- 18: **return** $\mathcal{F}_1$

them extremely sensitive to small variations in the design variables. In practice, this sensitivity is aggravated by the inherent variability of the physical properties of the materials and by the natural tolerances of manufacturing, cutting and assembly of the members. Under real conditions, a structure whose design lies exactly on the deterministic feasibility limit has a high probability of violating safety requirements or serviceability limit states if its final dimensions deviate even slightly from those specified. To mitigate this effect, the concept of robustness must be incorporated natively into the algorithmic process. The strategy adopted keeps the objective functions at the nominal design value, but also evaluates the feasibility of each candidate solution under a set of controlled perturbations.

The perturbation is applied exclusively to the girder diameter, d , because roundwood exhibits a natural diameter variability that is relevant to the structural behaviour. Instead of sampling the perturbation randomly, it follows a deterministic five-point grid, $\Lambda = \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$. This choice eliminates the seed bias that random sampling with few points would introduce, while the five levels suffice to capture possible feasibility steps within the interval $[-\rho, \rho]$ without relying on a monotonic response. For a given individual $\mathbf{x} = (d, b_w, h, esp_{long}, esp_{tab})$, $N_c = |\Lambda| = 5$ perturbed realizations, denoted by $\tilde{\mathbf{x}}^{(c)}$, are generated as defined in Eq. (4).

$$\tilde{d}^{(c)} = d(1 + \rho\lambda^{(c)}), \quad \lambda^{(c)} \in \Lambda, \quad \tilde{x}_i^{(c)} = x_i \text{ for } i \neq d \quad (4)$$

The term ρ represents the magnitude of the parametric deviation tolerated by the design with respect to the nominal diameter, and $\lambda^{(c)}$ runs over the five points of the grid Λ . When $\rho = 0$, the grid collapses to the single point $\lambda = 0$ and the procedure reduces to the standard deterministic evaluation ($N_c = 1$). In each cycle, the perturbed evaluations are processed independently in the analytical model, but only the constraints are re-evaluated at these points, while the objective functions remain computed at the nominal point $\mathbf{x}$, without any aggregation

over the grid. The constraint values of the N_c perturbed realizations are then aggregated by the worst-case (maximum) operator, according to Eq. (5).

$$g_j(\mathbf{x}) = \max_{c=1,\dots,N_c} g_j(\tilde{\mathbf{x}}^{(c)}) \quad (5)$$

With this formulation, a solution is deemed admissible only if all its constraints remain satisfied ($g_j \leq 0$) at every point of the diameter perturbation grid, that is, in the worst case. Designs set too close to the constraints violate at least one of the unfavourable grid points, which is enough for them to be penalized by the feasibility mechanism of the algorithm. The result is a natural shift of the Pareto front towards more conservative regions of the feasible design space.

3. Design of timber bridges

This section presents the design criteria adopted for roundwood log-girder timber bridges, covering the design vehicle actions, the partial factors for the mechanical properties, the deflection limits and the verification procedure of the girder and the deck.

3.1 Design vehicle load

The moving loads of Brazilian highway bridges are defined by NBR 7188 (2024) in its item 5.1.1. Since the bridges considered serve low-traffic rural roads, the TB-240 design vehicle is adopted. It is characterized by a 240 kN vehicle with six wheels and three axles, a load of 40 kN per wheel and an axle spacing of 1.5 m, complemented by a uniformly distributed crowd load of 4.0 kN/m² around it, as shown in Fig. 1.

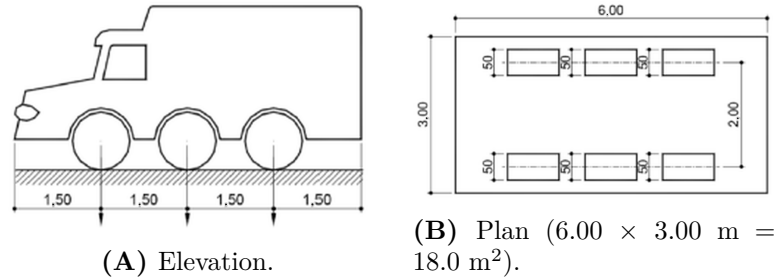


Figure 1: Configuration of the TB-240 design vehicle (NBR 7188, 2024).

3.2 Partial factors for mechanical properties

For the structural checks, NBR 7190-1 (2022) establishes strength (f) and modulus of elasticity (E) parameters for softwoods and hardwoods under the standard reference condition, that is, a moisture content (U) of 12%. The limit state method amplifies the internal forces and reduces these strengths by the modification factor (k_{mod}) and the material partial factor (γ_w), according to Eq. (6), where X_k is the characteristic strength and γ_w equals 1.4 for normal stresses and 1.8 for shear stresses.

$$X_d = \frac{k_{\text{mod}} \cdot X_k}{\gamma_w} \quad (6)$$

The modification factor is the product $k_{\text{mod},1} \cdot k_{\text{mod},2}$ (Eq. (7)), where the first factor depends on the load-duration class (item 5.8.4.1 of NBR 7190-1 (2022)) and the second on the moisture class (item 5.8.4.2).

$$k_{\text{mod}} = k_{\text{mod},1} \cdot k_{\text{mod},2} \quad (7)$$

NBR 7190-1 (2022) does not explicitly classify road traffic actions with respect to their cumulative duration. The classification of item 2.3.1.2 of Eurocode 5-2 (2004) and item NA.2.1 of the UK National Annex BSI (2004) is adopted here, which treat actions from vehicles and pedestrians as *short-term actions* owing to their transient and non-sustained nature, yielding $k_{\text{mod},1} = 0.90$ for solid timber. For the adopted moisture class (3), compatible with structural members exposed to the weather but protected from direct contact with water, a condition typical of roundwood bridges on rural roads, $k_{\text{mod},2} = 0.80$.

3.3 Deflection limits

3.3.1 Creep

The instantaneous deflections, $\delta_{\text{inst},G_i,k}$ and $\delta_{\text{inst},Q_j,k}$, associated with permanent and variable actions respectively, are computed with the mean modulus of elasticity in bending, $E_{0,m}$. The effect of time on stiffness is incorporated separately through the creep coefficient φ , which amplifies the instantaneous deflections to obtain the final deflections $\delta_{\text{fin},G_i,k}$ and $\delta_{\text{fin},Q_j,k}$, according to Eqs. (8) and (9).

$$\delta_{\text{fin},G_i,k} = \delta_{\text{inst},G_i,k} + \delta_{\text{creep}} = \delta_{\text{inst},G_i,k} \cdot (1 + \varphi) \quad (8)$$

$$\delta_{\text{fin},Q_j,k} = \delta_{\text{inst},Q_j,k} + \delta_{\text{creep}} = \delta_{\text{inst},Q_j,k} \cdot \psi_2 \cdot (1 + \varphi) \quad (9)$$

According to NBR 7190-1 (2022), timber has characteristics distinct from other construction materials, such as significant time-dependent deformation (creep). The creep coefficient φ (item 8.1, Table 20 of NBR 7190-1 (2022)) is defined as a function of the material and of the moisture class of the structure, increasing for higher moisture classes. For the moisture class (3) adopted in this work and for sawn timber or roundwood, the reference value is $\varphi = 0.8$.

3.3.2 Code deflection limits

For the serviceability limit state (SLS) check, item 8.2 of NBR 7190-1 (2022) establishes limit values for the instantaneous deflection (δ_{inst}) and the final deflection (δ_{fin}), which depend on the static configuration of the structure. For simply supported or continuous beams, which is the case of the structural system analysed in this work, the code interval lies between $L/300$ and $L/500$ for δ_{inst} and between $L/150$ and $L/300$ for δ_{fin} . The specific values adopted in this work are presented in Section 3.4 (Eq. (28)).

3.4 Bridge design

The structural system considered consists of a deck of rectangular timber planks, of width b_w and height h , continuously supported on circular roundwood girders of diameter d , analysed as simply supported beams of span L . The verification steps follow the sequence of determination of the internal forces, bending and shear checks of the girder, deflection check and, finally, bending check of the deck. The calculation procedure presented in this section follows the code criteria of NBR 7190-1 (2022) and the design procedure of the timber bridge design and construction manual by Calil Júnior et al. (2006).

3.4.1 Internal forces

For the solid circular girder and the rectangular deck plank, the area, second moment of area and section modulus of the cross-sections follow the usual geometric expressions for circular and rectangular sections, respectively.

The permanent actions correspond to the self-weight of the girders, the deck and other members, obtained from the geometry and the unit weight γ , according to Eq. (10).

$$G = \gamma \cdot A \quad (10)$$

The moving load considered is the TB-240 design vehicle presented in Section 3.1, represented by a three-axle, six-wheel vehicle, each wheel transmitting a load P of 40 kN, together with the crowd load, uniformly distributed over the occupation area with the characteristic value $q_A = 4.0$ kN/m². Since the girders are treated as isolated beams, the crowd load must be converted from an area load into a line load before entering Eqs. (18) and (21), by multiplying q_A by the tributary width of each girder, that is, by the corrected girder spacing $esp_{long,corr}$ (Section 4.3), according to Eq. (11).

$$q = q_A \cdot esp_{long,corr} \quad (11)$$

Dynamic effects are incorporated through the vertical impact coefficient (CIV), determined by Eqs. (12) and (13) according to item 5.1.3.1 of NBR 7188 (2024), where LIV is the characteristic length of the structure.

$$CIV = 1.35 \quad \text{for } L \leq 10.0 \text{ m} \quad (12)$$

$$CIV = 1 + 1.06 \left(\frac{20}{LIV + 50} \right), \quad \text{for } 10.0 < L \leq 200.0 \text{ m} \quad (13)$$

The CIV directly multiplies the internal forces due to variable actions.

The maximum bending moment and shear force due to the permanent load, as well as the corresponding static deflection, are obtained from classical statics of simply supported beams, according to Eqs. (14) and (15).

$$M_{gk} = \frac{g L^2}{8} \quad (14)$$

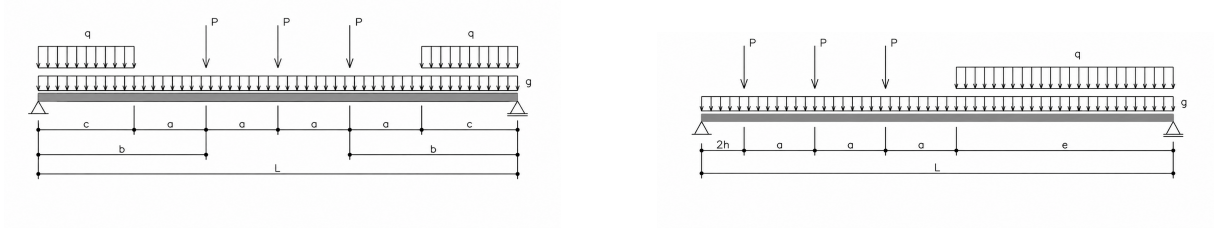
$$Q_{gk} = \frac{g L}{2} \quad (15)$$

The effects of the design vehicle are determined using structural analysis concepts and the recommendations of Calil Júnior et al. (2006).

Bending moment and deflection

For the bending moment and the deflection, the vehicle is centred on the span, with the central wheel located exactly at midspan and the two outer wheels at a distance a from it, as shown in Fig. 2A.

The maximum bending moment under the central wheel results from Eq. (17), written for $3 \text{ m} < L \leq 6 \text{ m}$. For spans greater than 6 m, Eq. (18) gives the maximum moment.



(A) Bending moment and deflection.

(B) Shear force.

Figure 2: Positioning of the design vehicle for computing the girder internal forces (Calil Júnior et al., 2006).

$$c = \frac{L - 4a}{2}, \quad \text{for } L > 6 \text{ m} \quad (16)$$

$$M_{qk} = \frac{3PL}{4} - Pa, \quad \text{for } 3 \text{ m} < L \leq 6 \text{ m} \quad (17)$$

$$M_{qk} = \left(\frac{3PL}{4} - Pa \right) + \frac{qc^2}{2}, \quad \text{for } L > 6 \text{ m} \quad (18)$$

The centred position, however, is the position of maximum moment only while the three wheels actually load the span. With the outer wheels at $L/2 \pm a$ from the left support, they reach the supports when $L = 2a$. From then on they have zero lever arm, or leave the span, and the centred arrangement no longer maximizes the moment. For $a = 1.5 \text{ m}$ and $L = 3.0 \text{ m}$, the outer wheels rest exactly on the supports and Eq. (17) degenerates into the case of a single centred load, $M_{qk} = PL/4$.

In this case, the maximum moment is found by positioning the design vehicle so that only two wheels lie within the span, following Barré's theorem, according to which the midspan must lie exactly halfway between one of the loads and the resultant of the pair. This condition leads to the position $x = (2L - a)/4$ for the first wheel, and the maximum moment is given by Eq. (19), obtained by maximizing $M(x) = Px(2L - 2x - a)/L$.

$$M_{qk}^{(2)} = \frac{P(2L - a)^2}{8L}, \quad \text{valid for } L \geq 1.5a \quad (19)$$

The envelope of the moment due to the moving load is therefore the largest value among the admissible arrangements, namely three centred wheels, two wheels in Barré's position, or a single wheel at midspan, according to Eq. (20).

$$M_{qk} = \max \left(\frac{3PL}{4} - Pa, \frac{P(2L - a)^2}{8L}, \frac{PL}{4} \right) \quad (20)$$

The first term governs for $L \geq 2.23a$, that is, $L \geq 3.34 \text{ m}$ when $a = 1.5 \text{ m}$, so that Eq. (20) recovers Eq. (17) over the usual range and differs from it only for the shortest spans. For the 3.0 m span, the centred formulation gives $30.0 \text{ kN}\cdot\text{m}$ against $33.75 \text{ kN}\cdot\text{m}$ from the envelope, an underestimation of 12.5%. For $L > 6 \text{ m}$, where the crowd load contribution appears, the centred arrangement governs by a wide margin and Eq. (18) remains governing.

The same centred position of Fig. 2A is used for the deflection (Section 4.3, Eqs. (25) and (27)), with b the distance between the support and the nearest outer wheel.

Shear force

For the shear force, Fig. 2B shows the vehicle placed at a distance $2h$ from the support, where h is the depth (or the diameter, for a circular section) of the member, the position at which NBR 7190-1 (2022) allows the shear force to be evaluated instead of at the face of the support itself. Since the diameter of the roundwood girder is not negligible compared with the span in the range of interest of this work, the design shear force Q_{qk} is obtained directly from Eq. (21), in which the operator $\max(\cdot, 0)$ cancels the contribution of the wheels and of the crowd load segment that exceed the finite length of the span.

$$Q_{qk} = \frac{P}{L} \left[\max(L - 2h, 0) + \max(L - 2h - a, 0) + \max(L - 2h - 2a, 0) \right] + \frac{q}{2L} \left[\max(L - 2h - 3a, 0) \right]^2 \quad (21)$$

The design internal forces, M_{sd} and Q_{sd} , follow the code load combination of NBR 8681 (2025). For a single-span bridge subjected to self-weight and to the design vehicle as the only leading variable action, this combination reduces to Eq. (22), where $\gamma_g = 1.30$ and $\gamma_q = 1.50$ are the partial factors for permanent and variable actions, respectively.

$$M_{sd} = \gamma_g M_{gk} + \gamma_q M_{qk}, \quad Q_{sd} = \gamma_g Q_{gk} + \gamma_q Q_{qk} \quad (22)$$

3.4.2 Bending design

The design bending strength, f_{md} , is obtained from the characteristic strength f_{mk} multiplied by the modification factor k_{mod} and divided by the partial factor γ_w , as already presented in Eq. (6). The bending check of the girder compares the design normal stress with this strength, according to Eq. (23).

$$\frac{\sigma_{Mx,d}}{f_{md}} = \frac{M_{sd}}{W_x f_{md}} \leq 1 \quad (23)$$

3.4.3 Shear design

For the solid circular section, the maximum shear stress occurs at the neutral axis and is compared with the design shear strength, $f_{v0,d}$, according to Eq. (24).

$$\tau_{\max} = \frac{4}{3} \frac{Q_{sd}}{A} \leq f_{v0,d} \quad (24)$$

3.4.4 Deflection design

The deflections due to the permanent load (δ_{gk}) and to the concentrated moving load (δ_{qk}) on the girder are obtained from Eqs. (25) and (27), where $E_{0,m}$ is the mean modulus of elasticity in bending and b is the distance between the support and the load application point, defined by Eq. (26).

$$\delta_{gk} = \frac{5 g L^4}{384 E_{0,m} I} \quad (25)$$

$$b = \frac{L - 2a}{2} \quad (26)$$

$$\delta_{qk} = \frac{P}{48 E_{0,m} I} \left[L^3 + 2b(3L^2 - 4b^2) \right] \quad (27)$$

These contributions correspond to the instantaneous deflections used in Eqs. (8) and (9), whose sum gives the total deflection δ_{fin} already including creep. The deflection check simultaneously compares this total deflection, in the quasi-permanent combination, with the code limit $L/350$, and the total instantaneous deflection, $\delta_{\text{inst}} = \delta_{gk} + \delta_{qk}$, due to permanent and variable actions in the characteristic (rare) combination without creep, with the limit $L/500$, both from Table 21 of NBR 7190-1 (2022), according to Eq. (28).

$$\max \left(\frac{\delta_{\text{fin}}}{L/350}, \frac{\delta_{gk} + \delta_{qk}}{L/500} \right) \leq 1 \quad (28)$$

3.4.5 Deck design

The deck plank is checked in isolation as a simply supported beam spanning the girder spacing ($esplong$). The bending moment due to the permanent load on the deck, $M_{gk,tab}$, follows the same formulation as Eq. (14), replacing L by $esplong$. The bending moment due to the concentrated load of one wheel of the design vehicle is obtained by placing a single wheel at the middle of this local span, as shown in Fig. 3, the critical position for an isolated concentrated load on a simply supported beam.

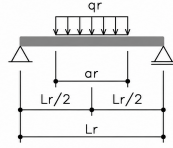


Figure 3: Critical position of a wheel on the deck (Calil Júnior et al., 2006).

$M_{qk,tab}$ is then obtained from Eq. (29), where a_r is the effective contact length of the wheel on the deck and CIV_{tab} is the vertical impact coefficient computed analogously to Eqs. (12) and (13), taking the spacing $esplong$ as the reference span.

$$M_{qk,tab} = \frac{P}{4} (esplong - a_r) \cdot CIV_{tab} \quad (29)$$

The design moment of the deck, $M_{sd,tab}$, follows the same load combination as Eq. (22). The bending check of the deck follows the same criterion used for the girder (Eq. (23)), comparing the design normal stress $\sigma_{x,d,tab} = M_{sd,tab}/W_{x,tab}$ with the design bending strength of the deck $f_{md,tab}$, computed according to Eq. (6).

4. Methodology

This section details the analysis and optimization framework developed for the parametric design of roundwood timber bridges, specifically of the girders and deck planks.

The framework was implemented in *Python* and made freely available through the *ReliaBridge* platform (ReliaBridge, 2026). Metaheuristic multi-objective optimization is carried out with the genetic algorithms of the *pymoo* library (Blank & Deb, 2020), while uncertainty quantification and stochastic global sensitivity analysis are performed with the *UQpy* library (Olivier et al., 2020).

For each processed solution, the platform provides the complete calculation report, indicating

the internal forces, the displacements and the outcome of each ultimate and serviceability code check.

The evaluation flow of each candidate solution follows the structure already presented in Algorithm 1 (Section 2.3). For each individual $\mathbf{x}$ generated by NSGA-II, the diameter is perturbed over a grid of N_c points and each realization is submitted to the **EvaluateModel** function. The objective functions remain computed at the nominal point $\mathbf{x}$, without any aggregation over the grid, whereas the constraints are aggregated by the worst-case operator (maximum, component-wise) over the N_c perturbed realizations, as described in Section 2.4. Algorithm 2 details the internal procedure of this function for a single design vector, nominal or perturbed.

Algorithm 2 EvaluateModel($\tilde{\mathbf{x}}$)

Require: Design vector $\tilde{\mathbf{x}} = (d, b_w, h, esp_{long}, esp_{tab})^T$

Ensure: Objective vector $\mathbf{f} = (f_1, f_2)$ and constraint vector $\mathbf{g} = (g_1, \dots, g_6)$

- 1: Run the geometric accommodation routine, determine n_{long} and n_{tab} and evaluate the spacing constraints g_5 and g_6
 - 2: Define the permanent and variable actions (design vehicle and crowd load) acting on the deck
 - 3: Distribute the deck reaction onto the girders (couples the loading of both members into a single girder loading)
 - 4: **Girder:** compute the internal forces (M_{sd}, Q_{sd}) and deflections, and evaluate g_1 (bending), g_2 (shear) and g_3 (deflection, with creep)
 - 5: **Deck:** compute the bending moment under the design vehicle wheel and evaluate g_4 (bending)
 - 6: Compute f_1 (total timber volume, Eq. (30)) and f_2 (deflection utilization, Eq. (31))
 - 7: **return** $\mathbf{f}, \mathbf{g}$
-

Step 3 highlights the coupling between deck and girder that characterizes the formulation. The deck reaction, arising from the permanent loads and the design vehicle, is distributed directly onto the girders at each evaluation, so that both members are designed simultaneously rather than in independent steps.

4.1 Test model definition and material properties

To test the tool and demonstrate the direct application of the implemented algorithm, a single-lane rural road bridge designed to carry the TB-240 design vehicle traffic specified by NBR 7188 (2024) was adopted as the case study. This configuration is representative of rural accesses and local roads, where the demand for timber designs is high.

The mechanical modelling of the structural members strictly follows the Brazilian code provisions. The material characteristics and the properties adopted in the simulations were established in accordance with NBR 7190-1 (2022). The elastic properties and the base strength classes for native forest species (D20 to D60), derived from tests on defect-free specimens, are listed in Table 2.

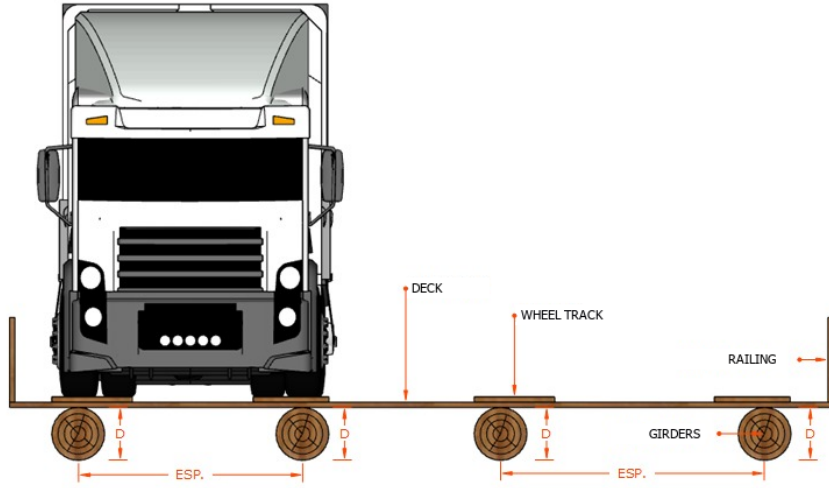
It should be noted that, according to item 6.3.4 of NBR 7190-1 (2022), when designing with the classes of Table 2 of the code, $f_{m,d} = f_{c0,d}$ is assumed, that is, the design bending strength is taken equal to the design compressive strength parallel to the grain. Since the modification factor and the partial factor for normal stresses (γ_{wf}) are the same for both checks, this equivalence translates, at the characteristic level, into $f_{m,k} = f_{c0,k}$. Hence, the bending strengths used by the platform in the limit state constraints are the tabulated $f_{c0,k}$ values of each class.

Table 2: Strength classes of native forest species and computed bending strength.

Class	$f_{c0,k}$ (MPa)	$f_{m,k}$ (MPa)	$f_{v0,k}$ (MPa)	$E_{c0,med}$ (MPa)	$\rho_{12\%}$ (kg/m ³)
D20	20	20.0	4	10 000	500
D30	30	30.0	5	12 000	625
D40	40	40.0	6	14 500	750
D50	50	50.0	7	16 500	850
D60	60	60.0	8	19 500	1 000

4.2 Structural system and parametric model

The structural system considered in the computational model consists of a deck made of longitudinal rectangular timber planks, continuously supported on longitudinal circular roundwood beams (girders), regularly distributed at a distance called the parametric spacing (esp). The deck is subjected to permanent actions from the self-weight of its constituent materials and to variable crowd and axle actions associated with moving traffic. The load transfer conveys the deck internal forces to the supporting girders, which are analysed as distributed loads along the span. The conceptual representation of the system is shown in Fig. 4.

**Figure 4:** Conceptual representation of the structural system of the timber bridge.

The model is mathematically parameterized. From a user input vector comprising the roadway width, mechanical properties, creep coefficients (φ) and code combination factors (ψ_2), the geometric properties of the cross-sections (area, second moment of area, section modulus and radii of gyration) are dynamically recomputed in the code for any generated individual. This parametric flexibility couples the geometry to the formulation of the geometric constraints, enabling feasibility checks in automated computational routines.

4.3 Constraint formulation and robust optimization

Within NSGA-II, the design is converted into a vector of mathematical variables $\mathbf{x} = (d, b_w, h, esp_{long}, esp_{tab})^T$, containing the girder diameter (d), the deck plank dimensions (b_w, h), the spacing between girders (esp_{long}) and the spacing between deck planks (esp_{tab}). From the spacings and the roadway width, the number of members of each family is determined internally by the geometric accommodation routine, which also gives rise to constraints g_5 and g_6 .

Both objective functions are to be minimized. The first, f_1 , is the total timber volume of the superstructure, according to Eq. (30), where n_{long} and n_{tab} are the numbers of girders and deck planks actually accommodated, A_{long} and A_{tab} are the respective cross-sectional areas and $b_{w,pista}$ is the roadway width. The second, f_2 , is the utilization of the girder serviceability deflection

limit, defined in Eq. (31) as the ratio between the total deflection, already including creep, and the code deflection limit $L/350$. Minimizing f_2 is equivalent to maximizing serviceability performance, and the two functions are conflicting, since reducing the timber volume necessarily increases the relative deflection.

$$f_1(\mathbf{x}) = V_{\text{total}} = n_{\text{long}} A_{\text{long}} L + n_{\text{tab}} A_{\text{tab}} b_{w,\text{pista}} \quad (30)$$

$$f_2(\mathbf{x}) = \frac{\delta_{\text{tot}}}{L/350} \quad (31)$$

The generated solutions must satisfy a strict set of six normalized constraint equations $g_j \leq 0$, namely girder bending stress (g_1), girder shear (g_2), girder deflection including creep (g_3), deck bending stress under the design vehicle wheels (g_4) and the code minimum and maximum spacing limits of the superstructure members (g_5 and g_6). The code procedure for computing the geometric properties, internal forces and the bending, shear and deflection checks of the girder, as well as the bending check of the deck, is detailed in Section 3.4.

Owing to the high natural variability of roundwood diameter, the platform introduces robustness by evaluating N_c realizations for each individual, perturbing only the girder diameter over a deterministic grid with deviation $\rho = 5\%$ (Section 2).

To measure the impact that the dimensional uncertainty of each of the five design variables, over its whole search domain, has on the six listed code constraints, the system integrates the computation of total-order Sobol sensitivity indices (S_{T_i}), evaluated by Monte Carlo simulation with the `UQpy` library (Olivier et al., 2020). This index numerically indicates to the designer which geometric characteristic of the bridge requires tighter quality control in cutting and field execution to prevent systemic failure of the deck.

5. Results and discussion

This section presents the results obtained with the optimization framework described in Section 4, applied to a matrix of roundwood timber bridge configurations for single-lane rural roads. The matrix combines four design spans with the five native-forest timber strength classes of NBR 7190-1 (2022), keeping the roadway width, the design vehicle and all code coefficients fixed, so that the observed differences arise exclusively from the span and the strength class.

The presentation is organized in four parts. The first describes the computational framework (Section 5.1) and defines the case matrix (Section 5.2). The second details a reference cell of the matrix (Section 5.3), in which all analytical features of the tool are exercised, namely hypervolume convergence, efficient front, utilization of the checks, cost of robustness, global Sobol sensitivity and dispersion of the design variables. The third covers the complete matrix (Section 5.4) and converts the results into preliminary design charts (Section 5.5). The fourth fits a closed-form pre-sizing equation to the database of the parametric sweep (Section 5.6), as a numerical alternative to the graphical reading of the charts.

5.1 Computational framework

The analysis and optimization framework described in Section 4 was implemented in an open-access computational platform for the design of timber bridges, capable of performing the complete verification of the structural members in accordance with NBR 7190-1 (2022) and NBR

7188 (2024) (ReliaBridge, 2026). For each processed solution, the platform provides the efficient front, the verification report and the descriptive statistics of the design variables, serving as an instrument for implementing and transferring the results of this work to the designer.

5.2 Definition of the case study matrix

The geometric, loading and code parameters kept constant across the whole matrix are presented in Table 3. The 4.5 m roadway width corresponds to the usual cross-section of a single-lane rural road bridge, compatible with one vehicle at a time and with side wheel guards.

Table 3: Geometric, loading and code parameters common to the whole matrix.

Parameter	Value
Roadway width, $b_{w,pista}$ (cm)	450
Design vehicle (ABNT NBR 7188)	TB-240
Wheel load, p_{rodak} (kN)	40
Distributed variable load, p_{qk} (kPa)	4
Distributed permanent load (wheel tracks), p_{gk} (kPa) ^a	0.10
Axle spacing, a (m)	1.5
Load-duration class	Short-term
Timber type	Natural timber
Moisture class	3
Partial factor for permanent actions, γ_g	1.30
Partial factor for variable actions, γ_q	1.50
Material partial factor (normal stresses), γ_{wf}	1.40
Material partial factor (shear), γ_{wc}	1.80
Quasi-permanent combination factor, ψ_2	0.30
Creep coefficient, φ	0.80
Robustness deviation, ρ (%)	5.0

^a The self-weight of the deck and girders is computed internally by the platform from the geometry and the timber density. The permanent load p_{gk} represents only the superimposed action of the wheel tracks, that is, the longitudinal running boards that guide the wheels. This contribution was estimated from two 5×30 cm boards with a density of 620 kg/m^3 ($\gamma \approx 6.1 \text{ kN/m}^3$), resulting in about 0.18 kN/m along the span, or approximately 0.04 kPa when distributed over the roadway width. A value of 0.10 kPa was adopted, rounded up to cover fastenings and the higher density usual for running boards.

The mechanical properties of the timber are not listed in this table because they vary among the cells of the matrix. They are those of Table 2, according to the strength class of each cell.

The case matrix is the Cartesian product of four design spans and five strength classes, totalling twenty configurations, as shown in Table 4. The span range of 3.0 to 6.0 m was chosen for two reasons. The first is representativeness, since it corresponds to the usual width of the watercourses crossed by roundwood bridges on rural roads. The second is formulation consistency, since within this interval the bending moment due to the moving load is given entirely by Eq. (17), without the additional crowd load term of Eq. (18), so that the result curves show no formulation discontinuity.

Table 4: Case study matrix. Each cell corresponds to an independent NSGA-II run.

Span L (m)	Strength class (ABNT NBR 7190-3)				
	D20	D30	D40	D50	D60
3.0	C-01	C-02	C-03	C-04	C-05
4.0	C-06	C-07	C-08	C-09	C-10
5.0	C-11	C-12	C-13	C-14	C-15
6.0	C-16	C-17	C-18	C-19	C-20

Cell C-13 ($L = 5.0$ m, class D40), highlighted in bold, is adopted as the reference case for the detailed analysis of Section 5.3.

The search bounds of the five design variables, common to all cells, are presented in Table 5. The bounds were set to cover the usual commercial dimensions of roundwood and sawn planks, without artificially restricting the search space of the algorithm.

Table 5: Bounds adopted for the design variables.

Variable	Minimum value (cm)	Maximum value (cm)
Girder diameter, d	20	100
Deck plank width, b_w	20	50
Deck plank height, h	5	15
Spacing between girders, esp_{long}	30	200
Spacing between deck planks, esp_{tab}	2	5

5.3 Detailed reference case: $L = 5.0$ m, class D40

This section exercises all analytical features of the platform on a single cell of the matrix, C-13. The choice falls on an intermediate span and an intermediate strength class, so that the observed behaviour is representative of the whole set. The results of the remaining nineteen cells are reported in condensed form in Section 5.4.

5.3.1 Convergence of the optimization

The quality of an approximated front is assessed, independently of visual inspection, by the hypervolume indicator (HV), which measures the portion of the objective space dominated by the front with respect to a reference point $\mathbf{r}$, according to Eq. (32) (Blank & Deb, 2020).

$$HV(\mathcal{F}_1, \mathbf{r}) = \text{Vol} \left(\bigcup_{\mathbf{x} \in \mathcal{F}_1} [f_1(\mathbf{x}), r_1] \times [f_2(\mathbf{x}), r_2] \right) \quad (32)$$

The objectives are normalized by the ideal and nadir points observed over the whole run, so that the indicator is dimensionless and comparable across cells of different scales, and the reference point $r_1 = r_2 = 1.1$ is adopted in this normalized space. Fig. 5 shows the evolution of the hypervolume with the number of generations for cell C-13.

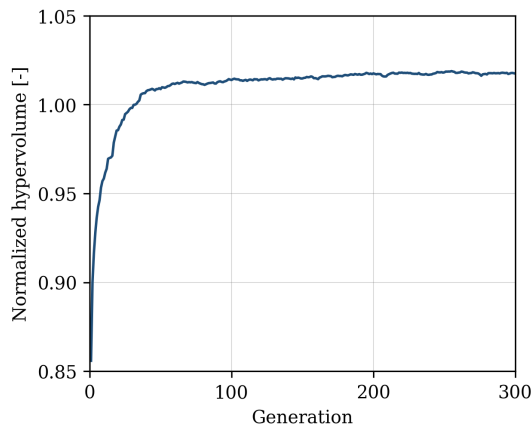


Figure 5: Evolution of the hypervolume over the NSGA-II generations.

The curve shows a steep increase in the first twenty generations and asymptotic refinement thereafter, stabilizing at generation 40, which confirms the convergence of the optimization with the number of generations adopted in Table 1. Across the twenty cells of the matrix, stabilization occurs between generations 21 and 153, always before the 300 generations executed. The spacing

of the front, which measures the variation of the distance between neighbouring solutions, was low over the whole matrix, between 0.0136 and 0.0203, which ensures a uniform distribution of the solutions along the front.

5.3.2 Efficient front and representative solutions

The robust multi-objective optimization of cell C-13, with $\rho = 5\%$, produced a set of non-dominated solutions distributed along the efficient front shown in Fig. 6. Table 6 lists a representative sample of these solutions, with the corresponding objective functions and the largest value among the six limit state functions, confirming compliance with the code criteria even under the robust evaluation.

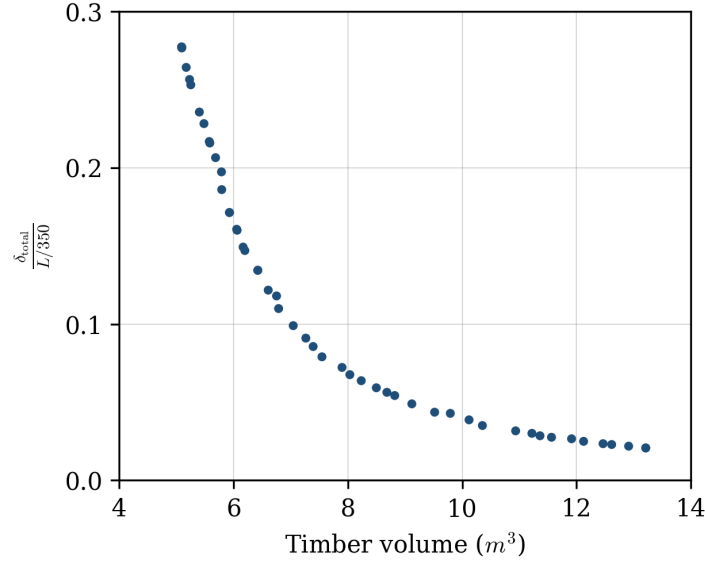


Figure 6: Efficient front of reference cell C-13 ($L = 5.0$ m, D40, $\rho = 5\%$).

Table 6: Representative solutions of the efficient front of cell C-13.

d (cm)	b_w (cm)	h (cm)	esp_{long} (cm)	esp_{tab} (cm)	V_{total} (m ³)	$\delta_{tot}/(L/350)$	g_{max}
47.78	45.51	7.34	69.98	2.24	5.09	0.2772	−0.0018
51.01	45.50	7.26	67.89	4.73	5.57	0.2167	−0.0431
56.41	45.50	11.29	110.54	4.69	6.06	0.1600	−0.0012
64.15	45.55	10.71	97.85	4.98	7.04	0.0990	−0.0471
75.71	45.50	10.09	92.98	2.55	8.82	0.0543	−0.0759
90.92	45.50	7.93	58.55	2.57	11.36	0.0285	−0.0959
100.00	45.50	6.99	81.23	4.53	13.21	0.0208	−0.0002

Seven solutions extracted from the final front of 50 non-dominated points, ordered from the minimum timber volume end (first row) to the minimum dimensionless deflection end (last row), including both ends. g_{max} is the largest value among the strength and serviceability limit state functions g_1 to g_4 of the solution, and it is negative in all rows, which confirms feasibility under the robust evaluation. The geometric limit state functions g_5 and g_6 , which only check that the spacings fall within the bounds of Table 5, are excluded from this maximum because they do not express a structural margin.

The front extends from 5.09 m³ of timber, with a total deflection equal to 27.7% of the code limit, to 13.21 m³, with a deflection of only 2.1% of the limit. The two objectives therefore vary on different scales. The volume slightly more than doubles (2.6 times), whereas the dimensionless deflection drops about thirteen times. This imbalance is a direct manifestation of the geometry of the circular section. Along the front, the volume grows approximately with $d^{1.28}$, whereas the deflection decreases with $d^{-3.54}$, so that the joint fit gives $\delta_{tot}/(L/350) \propto V^{-2.76}$, with a

coefficient of determination of 0.995. In design terms, this means that the first additional cubic metres of timber buy a substantial reduction in displacement. Doubling the volume from the economical end already reduces the deflection by about seven times, and doubling it again would reduce this value by a similar additional factor, although the studied front does not cover this second doubling.

The useful part of the front for the designer is therefore concentrated in the first half of the volume range. At 7.90 m^3 , about one third of the way between the ends, the deflection already drops to 7.2% of the limit. Beyond this point the marginal return falls, and going from $V = 8.82 \text{ m}^3$ to $V = 13.21 \text{ m}^3$ (an increase of 49.5%) reduces the deflection from only 5.4% to 2.1% of the limit, a gain of slightly more than three percentage points over an already ample margin. Increasing the volume beyond the first third of the front is therefore no longer justified in practice.

The active constraint along the front is the same from end to end. Deck bending (g_4) is the check closest to its limit in all 50 solutions of the front, without exception, including the minimum volume end, where it practically saturates at the code limit ($g_{\max} \approx 0$, utilization of 99.8%). Girder bending (g_1) follows closely at this same end, with a utilization of 99.7%, but does not overtake the deck as the governing constraint at any point of the front in this version of the matrix, which nevertheless confirms how close the two checks remain to each other. It can therefore be stated that moving along the front means increasing the girder diameter, which quickly gains margin, while the deck remains anchored at its limit throughout the whole trade-off space. The deck is thus not a secondary element of the design but the element that limits performance. The width-height pair of the plank and the girder spacing define the local bending span under the wheel, and they dictate the shape of the curve in Fig. 6.

5.3.3 Utilization of the code checks

Fig. 7 presents the utilization of each code check for a selected solution of the front, understood as the ratio between demand and resistance for the ultimate limit states, or between displacement and allowable limit for the serviceability limit state. A value of 100% corresponds to the code limit. Since the limit state functions are normalized in the form $g_j = (S_j - R_j)/R_j$, the utilization follows directly from them as $(1 + g_j) \times 100\%$. The solution analysed is the minimum timber volume solution of the front, the first row of Table 6, as it is of immediate interest for preliminary design.

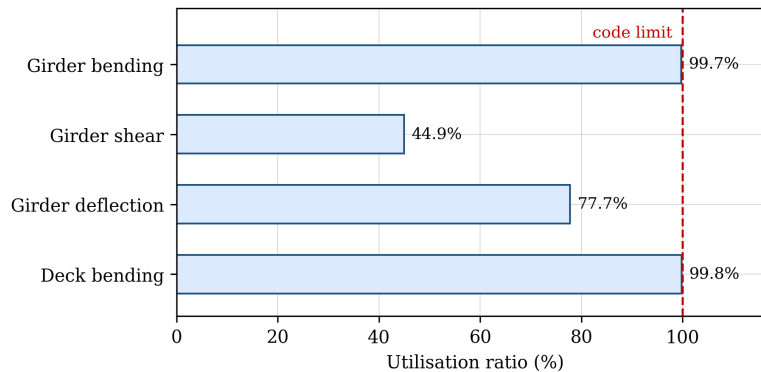


Figure 7: Utilization of each code check for the selected solution of the robust front of cell C-13.

The economical solution of cell C-13 shows two practically saturated checks and two with a

relevant margin. Deck bending governs the design, with a utilization of 99.8%, consistent with the $g_{\max} \approx 0$ recorded in the first row of Table 6. Immediately behind comes girder bending, at 99.7%. The two bending checks therefore become active almost simultaneously, which is the signature of a well-balanced design, since the algorithm could not reduce the volume further without violating one of them while the other was still at its limit.

The margins are concentrated in the remaining checks, although unevenly. Girder deflection (g_3 , which combines the quasi-permanent $L/350$ and characteristic $L/500$ limits) operates at 77.7% of its combined limit, and shear at only 44.9% of the design strength, the latter being the check with the largest reserve.

For the 5 m span, D40 timber and the TB-240 design vehicle, the design remains governed by bending strength rather than stiffness, but the deflection margin is no longer as wide as in the other cells. Section 5.4 shows that, for stiffer classes at the same span or for longer spans, this same check comes close to governing. The ample shear margin results from the circular geometry, in which the section area grows with d^2 while the shear demand is independent of the diameter, and indicates that the shear check, traditionally cited as critical for short roundwood members, is not decisive in this span range.

5.3.4 Cost of robustness

Deterministic optimization tends to place the optimal solutions exactly on the boundary of the active constraints, making them sensitive to small variations in the design variables. In roundwood timber bridges, this sensitivity is aggravated by the natural variability of the member diameter and by manufacturing and assembly tolerances. The incorporation of robustness shifts the efficient front into the interior of the feasible domain, penalizing solutions with a practically zero girder bending limit state function ($g_1 \approx 0$), since under a perturbation of $\pm\rho$ part of the realizations would violate it.

To quantify this effect, cell C-13 was re-optimized for five deviation levels on a uniform grid, $\rho \in \{0\%, 5\%, 10\%, 15\%, 20\%\}$, keeping all other algorithm parameters unchanged (Table 1). The case $\rho = 0\%$ corresponds to deterministic optimization, with $N_c = 1$, and serves as the reference. The grid was extended up to $\rho = 20\%$, well beyond the design value of 5% adopted in the rest of the matrix, to characterize the relationship between ρ and the volume cost over a wider range than strictly needed for design. The comparison is made at a single point of each front, the minimum volume solution (the end of immediate interest for preliminary design), rather than over paired ranges of dimensionless deflection, a reading discarded because it is sensitive to the noise of a single independent NSGA-II run per ρ level.

Table 7: Minimum volume of the efficient front of cell C-13 for each deviation level ρ , and percentage increase with respect to the deterministic case ($\rho = 0\%$).

ρ (%)	$V_{\min}$ (m ³)	Δ
0.0	4.800	–
5.0	5.090	+6.0%
10.0	5.414	+12.8%
15.0	5.788	+20.6%
20.0	6.091	+26.9%

$V_{\min}$ is the volume of the minimum timber volume solution of the efficient front at that ρ level. Δ is the percentage increase with respect to $V_{\min}(\rho=0\%)$.

Fig. 8 shows the five fronts shifting visibly and orderly into the interior of the feasible domain

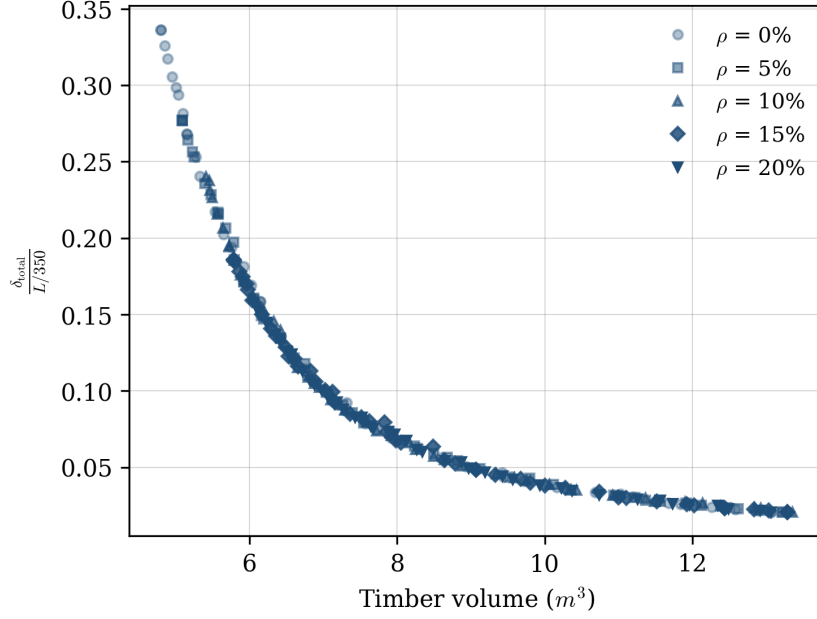


Figure 8: Efficient fronts of cell C-13 for $\rho = 0\%$, 5% , 10% , 15% and 20% .

as ρ increases. At the minimum volume end, Table 7 shows a monotonic increase approximately proportional to ρ , from 4.800 m^3 in the deterministic case to 5.090 m^3 (+6.0%) at $\rho = 5\%$, 5.414 m^3 (+12.8%) at $\rho = 10\%$, 5.788 m^3 (+20.6%) at $\rho = 15\%$ and 6.091 m^3 (+26.9%) at $\rho = 20\%$. The marginal cost rate increases slightly with ρ , from about 1.2% of additional volume per percentage point of ρ between 0 and 5% to about 1.3% to 1.4% per percentage point at the higher levels of the grid.

The value $\rho = 5\%$ remains the default across the whole matrix (Section 5.4) for consistency with the literature on dimensional tolerances of roundwood members.

5.3.5 Global sensitivity analysis by Sobol indices

To identify which design variables most influence the violation of each constraint under perturbation, a global sensitivity analysis was performed with the Sobol method (Sobol, 2001), as implemented in the UQpy library (Olivier et al., 2020). The total-effect indices S_{T_i} of the five design variables were estimated with respect to the six limit state functions, from an independent sample of $N = 20\,000$ points over the domain of each variable (Table 5). This sample size was defined from preliminary convergence tests, in which the variation of the Sobol indices was below 1% when the sample size was doubled.

Table 8: Most influential variable (dominant S_{T_i}) per constraint.

Constraint	Dominant variable	S_{T_i}	Share
g_1 : girder bending	d	1.005	98.2%
g_2 : girder shear	d	1.021	96.0%
g_3 : girder deflection	d	1.003	97.2%
g_4 : deck bending	esp_{long}	4.964	93.0%
g_5 : girder spacing	esp_{long}	1.003	100.0%
g_6 : deck plank spacing	esp_{tab}	1.000	100.0%

The relative share is the fraction of the sum of the total-effect indices of the five variables for that constraint, computed after truncating negative estimates at zero. The total-effect indices are estimated by variance differences and are therefore not bounded to the interval $[0, 1]$ for strongly non-linear responses. The value of 4.964 associated with g_4 reflects the non-linearity introduced by rounding the integer number of members accommodated on the roadway.

The result confirms the mechanical expectation for the three girder constraints. The diameter

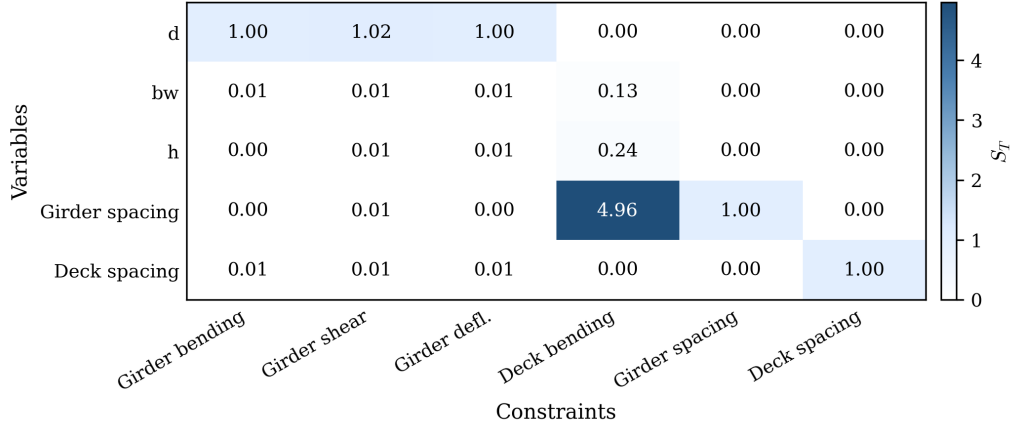


Figure 9: Total-effect Sobol indices (S_{T_i}) of the design variables on the constraints, cell C-13.

d accounts for 98.2% of the sensitivity of bending, 96.0% of shear and 97.2% of deflection, whereas the four remaining variables together account for less than 4% in any of them. This dominance is expected because d is the only variable that enters simultaneously the area, the section modulus and the second moment of area of the circular section, and it is even more pronounced in the serviceability limit state, in which the second moment of area varies with d^4 while the area, which governs the self-weight, varies only with d^2 .

Deck bending behaves differently and more informatively. In this check, the diameter is practically irrelevant, with a share of 0.1%, and the dominant variable becomes the girder spacing, with 93.0%, followed by the plank height, with 4.4%, and the plank width, with 2.5%. The girder spacing is the local bending span of the plank, in which the bending demand grows with the square of this span, whereas the plank resistance grows only with $b_w h^2$. The girder spacing is therefore the coupling variable between the two subsystems of the superstructure. It does not affect the girder in isolation, but defines how much load reaches each girder and which span the deck must bridge. The last two constraints are, by construction, exclusive functions of the respective spacings, and the analysis recovers them with a 100% share, which serves as a consistency check of the estimator itself.

A hierarchy for possible execution controls in timber bridges can be identified here. The diameter of the roundwood members and the girder spacing concentrate practically all the sensitivity of the system. These are the variables that justify investment in strict dimensional control, the former at the stage of log selection and grading, with rejection of members outside the nominal range, and the latter at the assembly stage, with a positioning template for the girders. The deck plank dimensions, b_w and h , and the deck plank spacing, esp_{tab} , admit looser execution tolerances without measurable loss of safety, since they account for a small portion of the variability of the limit state functions. This distinction is relevant for rural road construction, where quality control is necessarily selective and must be directed at what actually matters.

5.3.6 Dispersion of the design variables along the front

As a complement to the quality indicators of the front in the objective space, Fig. 10 characterizes the diversity of the front in the decision space, showing the dispersion of each design variable among the non-dominated solutions.

The dispersion is strongly uneven among the variables, both in absolute range and in relative variation. The girder spacing is by far the variable with the largest range along the front, with 62.2 cm of variation, a coefficient of variation of 20.4% and an interquartile range of 33.7 cm,

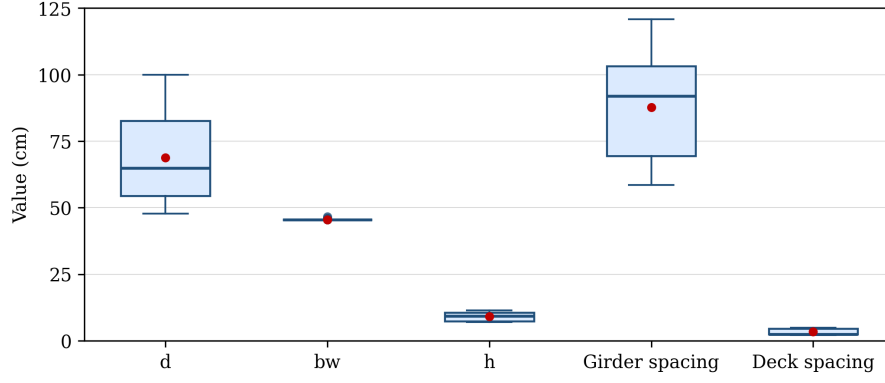


Figure 10: Dispersion of the design variables along the efficient front of cell C-13.

Table 9: Descriptive statistics of the design variables on the efficient front of cell C-13.

Variable	Mean	Std. dev.	CV (%)	Minimum	Q_1	Median	Q_3	Maximum	Range
d (cm)	68.84	17.14	24.90	47.78	54.49	64.89	82.72	100.00	52.22
b_w (cm)	45.58	0.20	0.44	45.50	45.50	45.50	45.51	46.60	1.10
h (cm)	9.13	1.64	17.92	6.99	7.34	9.36	10.68	11.39	4.40
esp_{long} (cm)	87.64	17.90	20.43	58.55	69.50	91.99	103.23	120.72	62.17
esp_{tab} (cm)	3.44	1.14	33.14	2.18	2.44	2.56	4.68	4.98	2.79

Statistics computed over the 50 non-dominated solutions of the final front.

covering a large part of the search domain of Table 5. Next comes the girder diameter, with a range of 52.2 cm and a coefficient of variation of 24.9%. These two variables are nevertheless the ones that build the trade-off between the two objectives, and no other variable comes close to them in absolute range.

At the opposite extreme, the deck plank width is practically invariant, with a coefficient of variation of 0.4% and an interquartile range of only 0.01 cm around 45.5 cm. The algorithm converges to this value regardless of the point of the front chosen by the designer, which has a direct practical consequence, since $b_w \approx 45.5$ cm can be fixed *a priori* in preliminary design without a performance penalty, reducing the effective dimensionality of the decision from five to four. The two remaining variables occupy intermediate positions, but with different readings. The plank height has a moderate coefficient of variation of 17.9% (4.4 cm range), whereas the deck plank spacing has 33.1% (2.8 cm range), much more dispersed, approaching the minimum admissible value of 2 cm at least at one point of the front (observed minimum of 2.18 cm, only 0.18 cm above the lower bound of the search domain).

The comparison with Section 5.3.5 is convergent. The two variables with the highest sensitivity are exactly the two with the largest absolute range on the front, and each one controls a subsystem, although their order is reversed depending on the indicator. The girder spacing is the variable with the largest range on the front and accounts for 93.0% of the sensitivity of deck bending, whereas the diameter, second in range, accounts for 96 to 98% of the sensitivity of the three girder checks. Moving along the front therefore consists of adjusting these two parameters simultaneously, one per subsystem. The three remaining variables, which combine low sensitivity and a small absolute range, behave as constructive parameters rather than decision variables. This suggests that a simplified preliminary design procedure could fix them and optimize only d and esp_{long} without significant loss of quality.

5.4 Parametric sweep: effect of span and strength class

This section reports the results of the twenty cells of the matrix. For each cell, the minimum timber volume solution was extracted from the efficient front, as it is of immediate interest for preliminary design, and the corresponding dimensions, total volume, dimensionless deflection and governing check were recorded. The consolidated results are given in Table 10.

Table 10: Consolidated results of the parametric sweep, minimum timber volume solution of each cell of the matrix.

Cell	L (m) / Class	d (cm)	b_w (cm)	h (cm)	esp_{long} (cm)	esp_{tab} (cm)	V_{total} (m ³)	$\delta_{tot}/(L/350)$	g_{max}	Governing check
C-01	3.0 / D20	43.30	38.59	6.41	49.35	4.24	2.988	0.0943	−0.0000	Deck bending
C-02	3.0 / D30	37.77	38.59	6.44	55.16	4.20	2.464	0.1364	−0.0000	Deck bending
C-03	3.0 / D40	34.50	39.44	6.05	59.94	4.92	2.154	0.1634	−0.0007	Deck bending
C-04	3.0 / D50	31.81	38.57	5.85	62.83	2.20	1.903	0.1992	−0.0000	Girder bending
C-05	3.0 / D60	30.12	38.58	5.57	64.70	5.00	1.745	0.2121	−0.0140	Deck bending
C-06	4.0 / D20	52.37	26.72	5.00	32.20	2.84	5.089	0.1355	−0.0008	Girder bending
C-07	4.0 / D30	45.85	45.73	8.71	72.78	2.69	4.076	0.1988	−0.0000	Deck bending
C-08	4.0 / D40	43.05	40.00	8.43	75.64	4.81	3.695	0.2149	−0.0068	Deck bending
C-09	4.0 / D50	38.71	35.69	5.04	54.90	4.81	3.163	0.2775	−0.0003	Deck bending
C-10	4.0 / D60	36.47	45.65	7.14	112.05	4.09	2.845	0.3132	−0.0004	Deck bending
C-11	5.0 / D20	60.13	45.51	8.07	54.32	4.23	7.331	0.1585	−0.0005	Girder bending
C-12	5.0 / D30	52.55	40.91	8.21	63.90	3.19	6.001	0.2277	−0.0001	Deck bending
C-13	5.0 / D40	47.78	45.51	7.34	69.98	2.24	5.090	0.2772	−0.0018	Deck bending
C-14	5.0 / D50	44.49	45.50	10.89	125.17	4.61	4.562	0.3543	−0.0004	Girder bending
C-15	5.0 / D60	41.79	45.50	6.55	78.07	3.01	4.084	0.3581	−0.0011	Deck bending
C-16	6.0 / D20	66.72	45.42	6.47	38.08	2.11	9.978	0.1740	−0.0002	Deck bending
C-17	6.0 / D30	58.17	45.42	12.74	106.87	4.84	7.907	0.2718	−0.0015	Girder bending
C-18	6.0 / D40	52.89	45.42	11.48	114.21	4.96	6.771	0.3333	−0.0003	Deck bending
C-19	6.0 / D50	49.24	45.43	10.55	119.44	2.45	6.015	0.3924	−0.0008	Deck bending
C-20	6.0 / D60	47.60	45.44	9.90	172.14	2.53	5.632	0.3914	−0.0000	Deck bending

All cells with $\rho = 5\%$ and the algorithm parameters of Table 1. The “Governing check” column records the constraint with the largest g_j value among g_1 to g_4 in the reported solution, that is, the one closest to the code limit, and g_{max} is the corresponding value.

Effect of span. Timber consumption grows more than proportionally with span in all classes, as expected, and the exponent of the power law $V \propto L^n$ fitted to the four observations of each class varies little among classes, between $n = 1.64$ (D40) and $n = 1.73$ (D20), with coefficients of determination of at least 0.996 in all cases. The general trend is a decrease of the exponent from D20 to D40 ($n = 1.73, 1.69$ and 1.64), followed by a partial recovery in the next two classes ($n = 1.67$ for D50 and $n = 1.68$ for D60), without strict monotonicity across the matrix. In absolute terms, going from 3.0 to 6.0 m multiplies consumption by 3.3 for class D20 and by 3.2 for class D60, a much narrower range than the differences between the governing checks of each class would suggest. The coupling between self-weight and performance, stronger in the lower strength classes, remains present as a mechanism, but its effect on the exponent is small in this span range.

Effect of strength class. In this matrix, the volume reduction obtained by moving up in class is approximately uniform across the four spans. Going from D20 to D60, consumption drops by 41.6% for the 3 m span, 44.1% for 4 m, 44.3% for 5 m and 43.6% for 6 m. Contrary to what a matrix with a more restricted search domain might suggest, none of the twenty cells saturates at the $d = 20$ cm lower bound of Table 5. Even the most economical cell, C-05 (3 m, D60), converges to $d = 30.1$ cm, above the bound. The table instead reveals a pattern of uniformly

tight structural convergence. In all twenty cells the minimum volume solution operates with at most 1.40% margin on the governing check ($g_{\max}$ between -0.0140 and -0.0000), with no relevant margin in any of them. The largest margin occurs in C-05 (3 m, D60) and the second largest in C-08 (4 m, D40, $g_{\max} = -0.0068$), with no clear systematic pattern between span and margin. This uniformity is the expected outcome of a well-converged algorithm exploring a continuous search domain, and it replaces the dimensional saturation effect that a narrower diameter range would produce.

For all spans, the first class upgrade is the most effective, by a considerable margin. Going from D20 to D30 reduces consumption by between 17.5% (3 m span) and 20.8% (6 m span), accounting alone for 41% to 48% of the total saving between D20 and D60, whereas the subsequent upgrades individually yield between 3.8% and 12.4% of the D20 volume. The flattening in the subsequent steps is caused by the trade-off identified in the problem formulation. Between D20 and D60 the characteristic bending strength triples and the modulus of elasticity almost doubles, but the density also doubles, from 500 to 1000 kg/m³, and the added self-weight consumes part of the strength gain. The gain neither vanishes nor reverses within the studied range, so that, strictly speaking, there is no optimal class in the sense of an interior minimum. From a preliminary design standpoint, the recommendation derived from the matrix is that specification efforts should concentrate on guaranteeing at least class D30, for any of the four spans, and that the pursuit of higher classes is justified only when high-class timber is available without a relevant cost premium.

The matrix is fully monotonic, which is the expected consistency check, since within each class the volume increases with span and within each span it decreases from class D20 to D60, without a single inversion among the twenty cells.

Governing check. Bending governs the entire matrix. Fifteen of the twenty cells are governed by deck bending and the remaining five by girder bending, with no cell governed by deflection. The two bending checks alternate across the matrix without a clear pattern in span or class, consistent with what Section 5.3.2 shows for the reference cell. The girder deflection check (g_3), although never governing, comes extremely close to its limit in three cells with the most severe span and class, with utilizations of 99.21% in C-15 ($L = 5.0$ m, D60), 99.90% in C-19 ($L = 6.0$ m, D50) and 99.97% in C-20 ($L = 6.0$ m, D60), always a few hundredths of a percentage point behind deck bending. The narrowest near-tie occurs in C-20, in which deck bending ($g_4 = -0.0000$, utilization of 100.00%) and girder deflection (99.97%) differ by less than three hundredths of a percentage point, and C-19 repeats the pattern, with the deck at 99.92% against deflection at 99.90%, decided by the fourth decimal place.

The predominance of the deck has a mechanical origin in the coupling through the girder spacing. Since the circular section becomes more volume-efficient as its diameter increases, the algorithm concentrates the material in a few girders and spreads them apart until the moment under the wheel, which grows with the local span of the plank (Eq. (29)), exhausts the capacity of the deck.

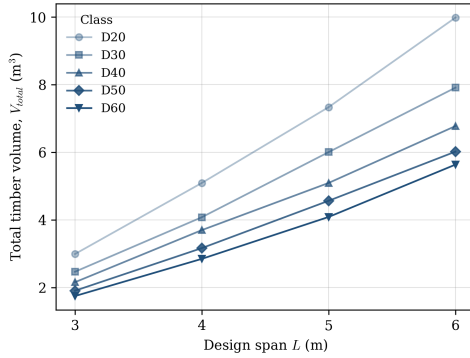
The practical consequence remains essentially the same, with a relevant caveat for the most severe classes and spans of the matrix. In roundwood timber bridges within this span range, under the TB-240 vehicle, bending strength rather than stiffness governs the entire matrix. The quasi-permanent index $\delta_{\text{tot}}/(L/350)$ remains below 39.3% in all twenty cells (maximum of

0.392 in C-19). The characteristic (rare) combination ($L/500$, without creep), however, already approaches its limit for the higher strength classes and longer spans, becoming nearly active in C-15, C-19 and C-20. This is a warning that, for spans or classes beyond those studied here, the transition from strength to stiffness governance may occur earlier than the isolated reading of the quasi-permanent index would suggest.

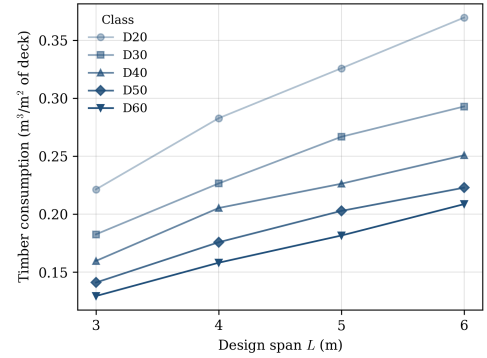
Shear, in turn, remains far from its limit throughout the matrix, with a maximum utilization of 60.2%. The area of the circular section grows with d^2 while the shear demand is independent of the diameter, so that this check, traditionally cited as critical for short roundwood members, does not approach its limit even after the correction of Eq. (21) for short spans.

5.5 Preliminary design charts

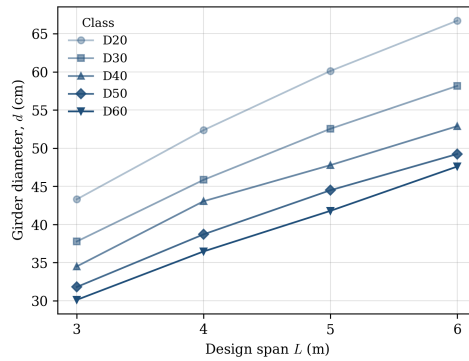
The results of Table 10 were converted into direct-reading design charts, intended for use by the designer at the conceptual stage, before any computational processing. Fig. 11A presents the total timber volume as a function of span, with one curve per strength class. Fig. 11B normalizes this volume by the deck area, allowing direct comparison between spans. Fig. 11C provides the required girder diameter, which is the most immediate procurement information for the engineering design.



(A) Total timber volume as a function of the design span.



(B) Timber consumption per square metre of deck.



(C) Required girder diameter as a function of the design span.

Figure 11: Preliminary design charts by strength class.

The choice of curve must correspond to the strength class of the timber actually available in the region of the site, not to the desired class, since local availability is the real constraint on rural roads. Fig. 11A gives the expected total volume, used for cost and transport logistics estimates, and Fig. 11C gives the required girder diameter, which defines the log size to be ordered. Fig. 11B normalizes consumption by the deck area to compare efficiency across spans,

and shows that the unit consumption varies from $0.129 \text{ m}^3/\text{m}^2$ for the 3 m span with D60 to $0.370 \text{ m}^3/\text{m}^2$ for the 6 m span with D20.

5.6 Pre-sizing equation

The twenty cells of Table 10 together form a database of twenty optimal designs labelled by span and strength class. Section 5.4 showed that, within each class, the volume follows a power law $V \propto L^n$ with $R^2 > 0.98$, but with a class-dependent exponent n , which prevents the direct use of a single curve outside the tabulated span-class pairs. A single multiplicative model, continuous in both span and strength class, was therefore fitted to the complete database in the form

$$V(L, f_{c0,k}) = a L^b f_{c0,k}^c, \quad (33)$$

where L is the design span in metres and $f_{c0,k}$ is the characteristic compressive strength parallel to the grain of each class, in MPa, taken from Table 2. The choice of $f_{c0,k}$ as the class regressor, rather than the modulus of elasticity $E_{c0,med}$, follows directly from the findings of Section 5.4, where bending strength, not stiffness, governs all twenty cells of the matrix. The parameters a , b and c were estimated by least squares on the linearized form $\ln V = \ln a + b \ln L + c \ln f_{c0,k}$, a fit that assigns equal weight to the relative error over the whole volume range of the matrix.

The fit gave $a = 2.290$, $b = 1.683$ and $c = -0.518$, with a coefficient of determination $R^2 = 0.999$ on the logarithmic scale. This value describes the quality of the fit over the same set of twenty cells used in the analysis and does not constitute an external validation of the model. Eq. (33) is a pre-sizing relationship (a surrogate model) fitted to the investigated domain, valid for spans of 3.0 to 6.0 m, strength classes D20 to D60, the single-lane structural configuration and the TB-240 design vehicle adopted in this work. Fig. 12 compares, for the twenty cells, the volume obtained by optimization with the volume estimated by Eq. (33). The points lie close to the diagonal $V = \hat{V}$ over the whole studied volume range, from 1.75 to 9.98 m^3 , without systematic deviation by span or class. The mean absolute relative error is 1.31%, with a maximum of 5.32% in cell C-08 ($L = 4.0 \text{ m}$, D40).

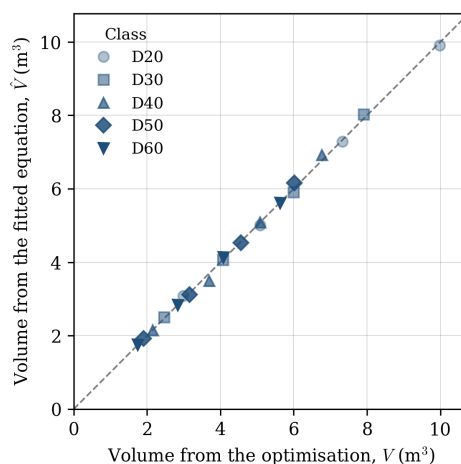


Figure 12: Volume estimated by the pre-sizing equation (Eq. (33)) against the volume obtained by optimization, for the twenty cells of the matrix.

The error is not concentrated in any identifiable subset of the matrix. Since Section 5.4 showed that none of the twenty cells saturates at the dimensional lower bound of the search domain, Eq. (33) does not suffer here from the limitation, common to fits of this kind, of

ignoring a side constraint of the algorithm. The maximum error of 5.32%, in cell C-08, reflects only the expected residual of a three-parameter fit over twenty observations, with no discernible relationship with the span or class of the cell. The signs of the fitted exponents have a direct physical reading and reinforce the consistency of the model with Section 5.4. The value $b = 1.683 > 1$ reproduces the same feedback between self-weight and section that produces exponents n between 1.64 and 1.73 per class, and $c = -0.518 < 0$ expresses that increasing $f_{c0,k}$ reduces the required volume, as expected. The exponent b of the joint equation indeed lies within the range of exponents n observed class by class, which indicates that tying the five curves into a single continuous surface in $f_{c0,k}$ does not distort the behaviour already identified for each class separately.

To verify the interpolation capability of Eq. (33) outside the fitting set, two additional configurations with intermediate spans not belonging to the matrix were optimized, $L = 4.5$ m with class D30 and $L = 5.5$ m with class D50, keeping all other parameters of Table 3 and of the algorithm (Table 1) unchanged. Neither of these two solutions took part in fitting the coefficients a , b and c . Table 11 compares the volume of the minimum volume solution obtained by optimization with the volume predicted by the equation.

Table 11: Verification of the pre-sizing equation at intermediate spans not used in the fit.

L (m)	Class	V optimization (m ³)	$\hat{V}$ equation (m ³)	Relative error
4.5	D30	4.969	4.944	-0.52%
5.5	D50	5.395	5.319	-1.42%

Optimizations with $\rho = 5\%$ and the same parameters as the twenty cells of the matrix. The relative error is $(\hat{V} - V)/V$.

The two errors, of -0.52% and -1.42%, are of the same order of magnitude as the mean error obtained over the fitting cells, and the optimized volumes lie between those of the neighbouring cells of the matrix with the same class (4.076 and 6.001 m³ for D30 between 4 and 5 m, and 4.562 and 6.015 m³ for D50 between 5 and 6 m). In both configurations girder bending is the governing check, in line with the behaviour observed in Section 5.4. This result indicates that the equation interpolates consistently between the tabulated spans, without, however, authorizing its use outside the investigated domain.

6. Conclusions

This work investigated, by means of a parametric robust multi-objective optimization framework in accordance with NBR 7190-1 (2022), how span, strength class and dimensional variability control material consumption, structural performance and the governing limit states of roundwood timber bridges on rural roads. The design is obtained without an initial assumption of cross-sections. The designer provides the span, roadway width, design vehicle and material properties, and receives the complete set of trade-off solutions, instead of checking a section previously chosen by experience or analogy. The framework covered the twenty cells of the matrix without manual intervention and without individual tuning of the algorithm parameters, with converged and uniformly distributed fronts in all of them, at a computational cost of about 2 min for the complete matrix on a personal computer.

The incorporation of robustness has a real but modest cost. In the reference cell, the minimum volume solution of the front became 6.0% more expensive at $\rho = 5\%$ with respect to the

deterministic case, and this increase grew monotonically up to 26.9% at $\rho = 20\%$, in exchange for protection against limit state violation under the natural dimensional deviation of roundwood.

Timber consumption grows with span according to a power law with exponent between 1.64 and 1.73, weakly dependent on the class, and the first class upgrade is the most effective. Moving from D20 to D30 reduces consumption by 17 to 21%, against 4 to 12% for each subsequent upgrade, because the increased density of the higher classes consumes part of the strength gain. These results were condensed into preliminary design charts and a pre-sizing equation $V(L, f_{c0,k}) = a L^b f_{c0,k}^c$, fitted with $R^2 = 0.999$ and a mean relative error of 1.3% within the investigated range of spans and classes, and made available through the open-access computational platform, which allow the timber volume and the girder diameter to be estimated without running a complete optimization.

In the 3 to 6 m range under the TB-240 vehicle, bending governs the entire matrix, fifteen cells through the deck and five through the girder, and shear remains far from its limit, with a maximum utilization of 60.2%. The deck is therefore not a secondary element of the design and deserves the same design attention as the girders. Girder deflection, however, already approaches its limit in the cells with the most severe span and class, a sign that the transition from strength to stiffness governance begins to emerge within the studied range itself.

The global sensitivity analysis showed that the diameter of the roundwood members and the girder spacing concentrate practically all the variability of the limit state functions, the former in the girder checks and the latter in deck bending. In rural road construction, where quality control is necessarily limited, this directs strict control to log selection and girder positioning, whereas the deck plank dimensions admit looser execution tolerances.

Limitations and future work

The scope is restricted to the superstructure of simply supported single-span bridges, under the TB-240 design vehicle and the code strength classes, without the design of connections, abutments and foundations. The robustness treated is geometric in nature, restricted to the girder diameter, and the cost of robustness was based on a single NSGA-II run per ρ level. Future work includes repeating these runs with multiple seeds, incorporating the variability of the mechanical properties and of the actions through reliability analyses (FORM or Monte Carlo simulation), using experimental properties of specific native species and validating the framework against the design of an actually built rural road bridge.

Declarations

Data and code availability

The computational platform described in this work is freely available (ReliaBridge, 2026). The data supporting the findings of this study are available from the corresponding author upon request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Wanderlei Malaquias Pereira Junior: Project administration, Methodology, Validation, Formal analysis, Writing (original draft), Writing (review and editing). **Priscilla Silva Teotônio:** Methodology, Software, Formal analysis, Writing (original draft). **Pedro Henrique Gomes Duarte:** Formal analysis, Writing (original draft). **Wellington Andrade da Silva:** Supervision, Validation. **André Luís Christoforo:** Methodology, Writing (review and editing). **João Paulo Machado Lopes:** Software. **Humberto Salazar Amorim Varum:** Writing (review and editing).

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September 24, 2026

To the Editor-in-Chief
Structures

Re: Submission of the manuscript “Robust multi-objective optimization of roundwood timber bridges: Structural performance and parametric design relationships”

Dear Editor,

We are pleased to submit the manuscript entitled “*Robust multi-objective optimization of roundwood timber bridges: Structural performance and parametric design relationships*” for consideration for publication in *Structures*.

The manuscript presents a parametric, robust multi-objective optimization framework for roundwood timber bridges, coupling NSGA-II to a structural model of the girder and deck and accounting for the natural dimensional variability of roundwood logs directly in the search. Applied to 20 configurations covering four spans and five strength classes, the framework reveals how span, strength class and dimensional variability control material efficiency and the governing limit states.

The main findings show that bending governs all 20 configurations, with the deck controlling 15 of them, while girder deflection already approaches its limit in the most severe cases. The first strength class upgrade (D20 to D30) delivers 41 to 48% of all the timber saving available up to D60, and robustness carries a real, monotonic cost of +6.0 to +26.9% in timber volume. These results are condensed into design charts and a closed-form pre-sizing equation ($R^2 = 0.999$) for direct use in preliminary design within the investigated domain.

To the best of our knowledge, this is the first study to integrate parametric modelling, multi-objective optimization and robustness analysis for roundwood timber bridges, a structural system still underexplored relative to concrete and steel bridges. We believe its combination of methodological novelty, structural insight and practical design tools fits well within the scope of *Structures*.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved its submission and declare no conflict of interest.

We appreciate your consideration and look forward to your response.

Sincerely, on behalf of all authors,

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Highlights

Robust multi-objective optimization of roundwood timber bridges: Structural performance and parametric design relationships

- Parametric robust multi-objective optimization framework for timber bridges.
- Robustness uses a deterministic diameter-perturbation grid, not random sampling.
- Timber consumption ranges from 1.75 to 9.98 m³ across 20 cases.
- Bending governs all 20 cells; deck bending controls 15 of them.
- Robustness cost is real: up to +26.9% extra volume at 20% deviation.